

using no agent on the fields. Thus, at the $\alpha = .05$ level, the biological agents are not effective in controlling weeds in the hay fields. However, the average hay production using the chemical agents appears to be greater than the hay production on fields with no weed agents.

When the sample sizes are not equal, the Dunnett procedure does not produce an experimentwise error rate equal to α . Dunnett (1964) provided adjustments to the values given in Table 11 for the unequal sample sizes.

9.8 Scheffé's *S* Method

The five multiple-comparison procedures discussed so far have been developed for pairwise comparisons among t population means. A more general procedure, proposed by Scheffé (1953), can be used to make all possible comparisons among the t population means. Although Scheffé's procedure can be applied to pairwise comparisons among the t population means, it is more conservative (less sensitive) than any of the other three multiple comparison procedures for detecting significant differences among pairs of population means because the "family" of comparisons it is trying to protect is larger than that for pairwise comparisons.

Scheffé's *S* Method for Multiple Comparisons

1. Consider any linear comparison among the t population means of the form

$$l = a_1\mu_1 + a_2\mu_2 + \cdots + a_t\mu_t$$

We wish to test the null hypothesis

$$H_0: l = 0$$

against the alternative

$$H_a: l \neq 0$$

2. The test statistic is

$$\hat{l} = a_1\bar{y}_1 + a_2\bar{y}_2 + \cdots + a_t\bar{y}_t$$

3. Let

$$S = \sqrt{\hat{V}(\hat{l})} \sqrt{(t-1)F_{\alpha, df_1, df_2}}$$

where, from Section 9.2,

$$\hat{V}(\hat{l}) = s_w^2 \sum_i \frac{a_i^2}{n_i}$$

t is the total number of population means, F_{α, df_1, df_2} is the upper-tail critical value of the F distribution with $a = \alpha$, $df_1 = t - 1$, and df_2 is the degrees of freedom for s_w^2 .

4. For a specified value of α , we reject H_0 if $|\hat{l}| > S$.
5. The error rate that is controlled is an *experimentwise error rate*. If we consider all imaginable contrasts, the probability of observing an experiment with one or more contrasts falsely declared to be significant is designated by α .

EXAMPLE 9.10

Refer to Example 9.3. We defined four contrasts in the $t = 5$ treatment means in an attempt to investigate the differences in the average hay production on fields treated with either the control or one of the four weed agents. Use the sample data and Scheffé's procedure to determine which if any of the four contrasts are significantly different from zero. Use $\alpha = .05$.

Solution The four contrasts of interest are given in Table 9.7 along with their estimates. To illustrate the calculations involved in Table 9.7, we will compute the value of S for the first contrast, control vs. agents. To compute

$$S = \sqrt{\hat{V}(\hat{l})} \sqrt{(t-1)F_{\alpha, df_1, df_2}}$$

we must first calculate $\hat{V}(\hat{l})$. Using the formula

$$\hat{V}(\hat{l}) = s_W^2 \sum_{i=1}^t \frac{a_i^2}{n_i}$$

with all samples sizes equal to 6 and $s_W^2 = .0153$, we have

$$\hat{V}(\hat{l}) = .0153 \left(\frac{(4)^2}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} + \frac{1}{6} \right) = .0153 \frac{20}{6} = .0510$$

TABLE 9.7 Computations for Scheffé procedure in weed control experiment

Contrast	Treatment					$\sum a_i^2/n_i$	$\hat{l}$	$\hat{V}(\hat{l})$	S	Conclusion
	Control a_1	Bio1 a_2	Bio2 a_3	Chm1 a_4	Chm2 a_5					
Control vs. agents	4	-1	-1	-1	-1	20/6	-.836	.0510	.750	Significant
Biological vs. chemical	0	1	1	-1	-1	4/6	-.294	.0102	.336	Not significant
Biol vs. Bio2	0	1	-1	0	0	2/6	-.035	.0051	.237	Not significant
Chm1 vs. Chm2	0	0	0	1	-1	2/6	-.085	.0051	.237	Not significant

From Table 8 for $\alpha = .05$, $df_1 = t - 1 = 4$, $df_2 = 25$ (the degrees of freedom for s_W^2), $F_{.05, 4, 25} = 2.76$. The computed value of S is then

$$S = \sqrt{.0510} \sqrt{4(2.76)} = (.2258)(3.323) = .750$$

Because the absolute value of $\hat{l}$ is $|- .836| = .836$ exceeds .750, we have significant evidence ($\alpha = .05$) to indicate that the average hay production from the fields treated with a weed agent exceeds the average yield in the fields having no treatment for weeds. The calculations for the other three contrasts are summarized in Table 9.7. Note that the value of S changes for the different contrasts. In our example, the only contrast significantly different from zero was the first contrast. The remaining three contrasts were not significant at the $\alpha = .05$ level. These conclusions are different from the conclusions we reached in Example 9.4, where we found that the second contrast was also significantly different from zero. The reason for the differences in conclusions is that the Scheffé procedure controls the experimentwise Type I error rate at level .05, whereas in Example 9.4 we only control the individual comparison rate at level .05.

Scheffé's confidence interval

Scheffé's method can also be used for constructing a simultaneous confidence interval for all possible (not necessarily pairwise) contrasts using the t treatment